

1. Analytical Derivation of Sphere-of-Influence Encounter Points

1.1. Problem Statement

Given two Keplerian elliptical orbits (spacecraft orbit 1, planet orbit 2) in heliocentric ecliptic coordinates, find the eccentric anomalies E_1, E_2 such that the distance between the spacecraft and the planet equals the planet's sphere-of-influence radius r_{SOI} :

$$\|A\mathbf{r}_1(E_1) - B\mathbf{r}_2(E_2)\| = r_{\text{SOI}} \quad (1)$$

where A and B are known constant 3×2 rotation matrices mapping from orbital-plane 2D coordinates to 3D heliocentric ecliptic coordinates.

1.2. Notation

- a_i, b_i, e_i : semi-major axis, semi-minor axis, and eccentricity of orbit i (with $b_i = a_i \sqrt{1 - e_i^2}$)
- E_i : eccentric anomaly of orbit i
- $A \in \mathbb{R}^{3 \times 2}, B \in \mathbb{R}^{3 \times 2}$: orbital-to-ecliptic rotation matrices with components A_{jk}, B_{jk}
- r_{SOI} : radius of the planet's sphere of influence

1.3. Position in Orbital Plane

For a Keplerian ellipse, the 2D position vector in the orbital plane as a function of eccentric anomaly is:

$$\mathbf{r}_i(E_i) = \begin{pmatrix} a_i(\cos E_i - e_i) \\ b_i \sin E_i \end{pmatrix} \quad (2)$$

1.4. Distance Constraint

Square both sides of the constraint:

$$(A\mathbf{r}_1 - B\mathbf{r}_2)^\top (A\mathbf{r}_1 - B\mathbf{r}_2) = r_{\text{SOI}}^2 \quad (3)$$

Expand the left-hand side:

$$\mathbf{r}_1^\top A^\top A \mathbf{r}_1 - 2\mathbf{r}_1^\top A^\top B \mathbf{r}_2 + \mathbf{r}_2^\top B^\top B \mathbf{r}_2 = r_{\text{SOI}}^2 \quad (4)$$

1.4.1. Orthonormality of A and B

Since A and B are 3×2 matrices whose columns are orthonormal (they embed an orthonormal frame from the orbital plane into 3D space):

$$A^\top A = I_{2 \times 2}, \quad B^\top B = I_{2 \times 2} \quad (5)$$

Therefore the self-terms simplify:

$$\mathbf{r}_1^\top A^\top A \mathbf{r}_1 = \mathbf{r}_1^\top \mathbf{r}_1 = |\mathbf{r}_1|^2 =: r_1^2 \quad (6)$$

$$\mathbf{r}_2^\top B^\top B \mathbf{r}_2 = \mathbf{r}_2^\top \mathbf{r}_2 = |\mathbf{r}_2|^2 =: r_2^2 \quad (7)$$

1.4.2. The coupling matrix C

Define the 2×2 matrix:

$$C := 2A^\top B \quad (8)$$

This encodes the entire mutual orientation of the two orbital planes. Equation 4 becomes:

$$r_1^2 + r_2^2 - \mathbf{r}_1^\top C \mathbf{r}_2 = r_{\text{SOI}}^2 \quad (9)$$

Note: C is **not** necessarily symmetric — it is a general 2×2 real matrix with $|C_{jk}| \leq 2$.

1.5. Newton's Method

We seek to minimise $f(E_1, E_2) := r_1^2 + r_2^2 - \mathbf{r}_1^\top C \mathbf{r}_2 - r_{\text{SOI}}^2$ using Newton's method. Each iteration solves the 2×2 linear system $H\delta = -\nabla f$ for the step $\delta = (\delta E_1, \delta E_2)^\top$, then updates $E_i \leftarrow E_i + \delta E_i$.

1.5.1. Deriving the gradient and Hessian

The objective is $f = r_1^2 + r_2^2 - \mathbf{r}_1^\top \mathbf{C} \mathbf{r}_2 - r_{\text{SOI}}^2$, with r_{SOI}^2 constant. We differentiate each part with respect to E_i .

1.5.1.1. Self-terms r_i^2

Using $r_i = a_i(1 - e_i \cos E_i)$:

$$\frac{\partial r_i^2}{\partial E_i} = 2a_i^2 e_i \sin E_i (1 - e_i \cos E_i) \quad (10)$$

$$\frac{\partial^2 r_i^2}{\partial E_i^2} = 2a_i^2 e_i (\cos E_i - e_i + 2e_i \sin^2 E_i) \quad (11)$$

and $\partial r_i^2 / \partial E_j = 0$ for $i \neq j$.

1.5.1.2. Cross-term $\mathbf{r}_1^\top \mathbf{C} \mathbf{r}_2$

The position vector Equation 2 can be factored as $\mathbf{r}_i = \text{diag}(a_i, b_i) \mathbf{p}_i$, where $\text{diag}(a_i, b_i) := \begin{pmatrix} a_i & 0 \\ 0 & b_i \end{pmatrix}$ is a diagonal matrix and $\mathbf{p}_i$ is a dimensionless position-direction vector:

$$\mathbf{p}_i = \begin{pmatrix} \cos E_i - e_i \\ \sin E_i \end{pmatrix} \quad (12)$$

Substituting $\mathbf{r}_i = \text{diag}(a_i, b_i) \mathbf{p}_i$ into the cross-term and using $(\mathbf{X} \mathbf{p})^\top = \mathbf{p}^\top \mathbf{X}^\top$:

$$\mathbf{r}_1^\top \mathbf{C} \mathbf{r}_2 = (\text{diag}(a_1, b_1) \mathbf{p}_1)^\top \mathbf{C} \text{diag}(a_2, b_2) \mathbf{p}_2 = \mathbf{p}_1^\top \text{diag}(a_1, b_1)^\top \mathbf{C} \text{diag}(a_2, b_2) \mathbf{p}_2 \quad (13)$$

Since diagonal matrices are symmetric ($\text{diag}(x, y)^\top = \text{diag}(x, y)$):

$$\mathbf{r}_1^\top \mathbf{C} \mathbf{r}_2 = \mathbf{p}_1^\top \text{diag}(a_1, b_1) \mathbf{C} \text{diag}(a_2, b_2) \mathbf{p}_2 \quad (14)$$

We have a chain of five matrices. Applying associativity ($\mathbf{A}(\mathbf{B}\mathbf{C}) = (\mathbf{A}\mathbf{B})\mathbf{C}$) twice – first to $(\mathbf{p}_1^\top \cdot \text{diag}(a_1, b_1)) \cdot \mathbf{C} = \mathbf{p}_1^\top \cdot (\text{diag}(a_1, b_1) \cdot \mathbf{C})$, then to $(\mathbf{p}_1^\top \cdot (\text{diag}(a_1, b_1) \cdot \mathbf{C})) \cdot \text{diag}(a_2, b_2) = \mathbf{p}_1^\top \cdot ((\text{diag}(a_1, b_1) \cdot \mathbf{C}) \cdot \text{diag}(a_2, b_2))$ – lets us group the three constant middle factors into a single matrix. We define the **scaled coupling matrix**:

$$\mathbf{M} := \text{diag}(a_1, b_1) \mathbf{C} \text{diag}(a_2, b_2) = \begin{pmatrix} C_{11}a_1a_2 & C_{12}a_1b_2 \\ C_{21}a_2b_1 & C_{22}b_1b_2 \end{pmatrix} \quad (15)$$

giving $\mathbf{r}_1^\top \mathbf{C} \mathbf{r}_2 = \mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2$.

Since $\mathbf{M}$ is constant, differentiation acts only on $\mathbf{p}_1$ and $\mathbf{p}_2$. Define the derivative vectors:

$$\hat{\mathbf{w}}_i := \frac{d\mathbf{p}_i}{dE_i} = \begin{pmatrix} -\sin E_i \\ \cos E_i \end{pmatrix}, \quad \hat{\mathbf{u}}_i := \begin{pmatrix} \cos E_i \\ \sin E_i \end{pmatrix} \quad (16)$$

To differentiate $\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2$ with respect to E_1 , note that only $\mathbf{p}_1$ depends on E_1 . By associativity, $\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2 = \mathbf{p}_1^\top (\mathbf{M} \mathbf{p}_2)$, which is a dot product of $\mathbf{p}_1$ with the constant vector $\mathbf{M} \mathbf{p}_2$. The derivative of a dot product with one constant factor is:

$$\frac{\partial(\mathbf{p}_1^\top (\mathbf{M} \mathbf{p}_2))}{\partial E_1} = \hat{\mathbf{w}}_1^\top (\mathbf{M} \mathbf{p}_2) = \hat{\mathbf{w}}_1^\top \mathbf{M} \mathbf{p}_2 \quad (17)$$

Similarly, for E_2 only $\mathbf{p}_2$ depends on E_2 . By associativity, $\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2 = (\mathbf{p}_1^\top \mathbf{M}) \mathbf{p}_2$, a dot product of the constant row vector $\mathbf{p}_1^\top \mathbf{M}$ with $\mathbf{p}_2$:

$$\frac{\partial((\mathbf{p}_1^\top \mathbf{M}) \mathbf{p}_2)}{\partial E_2} = (\mathbf{p}_1^\top \mathbf{M}) \hat{\mathbf{w}}_2 = \mathbf{p}_1^\top \mathbf{M} \hat{\mathbf{w}}_2 \quad (18)$$

The same reasoning applies to higher derivatives, replacing $\mathbf{p}_i$ by successive derivatives $\hat{\mathbf{w}}_i, -\hat{\mathbf{u}}_i$:

$$\frac{\partial^2(\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2)}{\partial E_1^2} = -\hat{\mathbf{u}}_1^\top \mathbf{M} \mathbf{p}_2, \quad \frac{\partial^2(\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2)}{\partial E_2^2} = -\mathbf{p}_1^\top \mathbf{M} \hat{\mathbf{u}}_2 \quad (19)$$

$$\frac{\partial^2(\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2)}{\partial E_1 \partial E_2} = \hat{\mathbf{w}}_1^\top \mathbf{M} \hat{\mathbf{w}}_2 \quad (20)$$

1.5.1.3. Combined expressions

Combining $f = r_1^2 + r_2^2 - \mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2 - r_{\text{SOI}}^2$ using Equation 10 with Equation 17, and Equation 11 with Equation 19 and Equation 20:

1.5.2. Gradient

$$f_1 = 2a_1^2 e_1 \sin E_1 (1 - e_1 \cos E_1) - \hat{\mathbf{w}}_1^\top \mathbf{M} \mathbf{p}_2 \quad (21)$$

$$f_2 = 2a_2^2 e_2 \sin E_2 (1 - e_2 \cos E_2) - \mathbf{p}_1^\top \mathbf{M} \hat{\mathbf{w}}_2 \quad (22)$$

1.5.3. Hessian

The Hessian $\mathbf{H}$ is symmetric ($H_{12} = H_{21}$):

$$H_{11} = 2a_1^2 e_1 (\cos E_1 - e_1 + 2e_1 \sin^2 E_1) + \hat{\mathbf{u}}_1^\top \mathbf{M} \mathbf{p}_2 \quad (23)$$

$$H_{22} = 2a_2^2 e_2 (\cos E_2 - e_2 + 2e_2 \sin^2 E_2) + \mathbf{p}_1^\top \mathbf{M} \hat{\mathbf{u}}_2 \quad (24)$$

$$H_{12} = -\hat{\mathbf{w}}_1^\top \mathbf{M} \hat{\mathbf{w}}_2 \quad (25)$$

1.5.4. Solving the Newton step

The 2×2 system is solved directly via Cramer's rule:

$$\Delta = H_{11}H_{22} - H_{12}^2 \quad (26)$$

$$\delta E_1 = \frac{f_1 H_{22} - f_2 H_{12}}{-\Delta}, \quad \delta E_2 = \frac{f_2 H_{11} - f_1 H_{12}}{-\Delta} \quad (27)$$

All coupling terms are bilinear forms $\mathbf{x}^\top \mathbf{M} \mathbf{y}$ with 2D vectors, each requiring 4 multiplies and 3 adds. The five needed dot products ($\hat{\mathbf{w}}_1^\top \mathbf{M} \mathbf{p}_2$, $\mathbf{p}_1^\top \mathbf{M} \hat{\mathbf{w}}_2$, $\hat{\mathbf{u}}_1^\top \mathbf{M} \mathbf{p}_2$, $\mathbf{p}_1^\top \mathbf{M} \hat{\mathbf{u}}_2$, $\hat{\mathbf{w}}_1^\top \mathbf{M} \hat{\mathbf{w}}_2$) can share the intermediate products $\mathbf{M} \mathbf{p}_2$, $\mathbf{M} \hat{\mathbf{w}}_2$, $\mathbf{M} \hat{\mathbf{u}}_2$ (each a 2D matrix-vector multiply).